HTML / Web Page / Form Design

1. **Design a webpage for the following layout For Student profile – where student roll number, name, contact, photo, class and area of interest in column 2. In Colum 1 provide the hyperlinks for Home, Contact us and about us**

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Profile</title>
    <style>
        body {
            font-family: Arial, sans-serif;
        }
        .container {
            display: flex;
            width: 80%;
            margin: auto;
            border: 1px solid black;
        }
        .column1 {
            width: 25%;
            background-color: #f2f2f2;
            padding: 20px;
        }
        .column2 {
```

```
            width: 75%;
            padding: 20px;
        }
        a {
            display: block;
            margin: 10px 0;
            text-decoration: none;
            color: blue;
            font-weight: bold;
        }
        img {
            width: 150px;
            height: 150px;
            border: 1px solid black;
        }
    </style>
</head>
<body>
<h2 align="center">Student Profile Page</h2>
<div class="container">
    <!-- Column 1 -->
    <div class="column1">
        <h3>Menu</h3>
```

```
        <a href="#">Home</a>
        <a href="#">Contact Us</a>
        <a href="#">About Us</a>
    </div>
    <!-- Column 2 -->
    <div class="column2">
        <h3>Student Details</h3>
        <p><b>Roll Number:</b> 101</p>
        <p><b>Name:</b> Nikita</p>
        <p><b>Contact:</b> 9876543210</p>
        <p><b>Class:</b> TYBSc IT</p>
        <p><b>Area of Interest:</b> Web Development</p>
        <p><b>Photo:</b></p>
        <img src="student.jpg" alt="Student Photo">
    </div>
</div>
</body>
</html>
```

2. **Create a login form with a username and password. Display “Welcome” message if username and password is same otherwise display “Invalid username or password” message.**

```
<!DOCTYPE html>
<html>
```

```
<head>
    <title>Login Form</title>
    <script>
        function checkLogin() {
            var username = document.getElementById("username").value;
            var password = document.getElementById("password").value;

            if (username === "admin" && password === "1234") {
                alert("Welcome");
            } else {
                alert("Invalid username or password");
            }
        }
    </script>
</head>
<body>
<h2>Login Form</h2>
<form onsubmit="checkLogin(); return false;">
    Username:<br>
    <input type="text" id="username" required><br><br>
    Password:<br>
    <input type="password" id="password" required><br><br>
    <input type="submit" value="Login">
```

```
</form>

</body>

</html>
```

3. **Create an html page with 7 separate lines in different colors. State color of each line in its text.**

```
<!DOCTYPE html>

<html>

<head>

   <title>Seven Colored Lines</title>

</head>

<body>

<h2>Seven Colored Lines</h2>

<p style="color:red;">This line is Red color</p>

<p style="color:blue;">This line is Blue color</p>

<p style="color:green;">This line is Green color</p>

<p style="color:orange;">This line is Orange color</p>

<p style="color:purple;">This line is Purple color</p>

<p style="color:brown;">This line is Brown color</p>

<p style="color:black;">This line is Black color</p>

</body>

</html>
```

4. **Design a HTML form for Bus Ticket Reservation consisting of fields for Name, Address, contact no, source station(Dropdown list), Destination station, Date of booking, date of journey, no of passenger, name of passenger, gender of passenger etc. (Use proper alignment)**

```html
<!DOCTYPE html>
<html>
<head>
    <title>Bus Ticket Reservation</title>
    <style>
        body {
            font-family: Arial, sans-serif;
        }
        table {
            margin: auto;
            border-collapse: collapse;
        }
        td {
            padding: 8px;
        }
        h2 {
            text-align: center;
        }
    </style>
</head>
<body>
<h2>Bus Ticket Reservation Form</h2>
<form>
```

```
<table border="1">
    <tr>
        <td>Name</td>
        <td><input type="text" name="name"></td>
    </tr>
    <tr>
        <td>Address</td>
        <td><textarea name="address" rows="3" cols="20"></textarea></td>
    </tr>
    <tr>
        <td>Contact No</td>
        <td><input type="tel" name="contact"></td>
    </tr>
    <tr>
        <td>Source Station</td>
        <td>
            <select name="source">
                <option>Select</option>
                <option>Mumbai</option>
                <option>Pune</option>
                <option>Nashik</option>
                <option>Nagpur</option>
            </select>
```

```
    </td>
</tr>
<tr>
    <td>Destination Station</td>
    <td>
        <select name="destination">
            <option>Select</option>
            <option>Mumbai</option>
            <option>Pune</option>
            <option>Nashik</option>
            <option>Nagpur</option>
        </select>
    </td>
</tr>
<tr>
    <td>Date of Booking</td>
    <td><input type="date" name="booking_date"></td>
</tr>
<tr>
    <td>Date of Journey</td>
    <td><input type="date" name="journey_date"></td>
</tr>
<tr>
```

```
        <td>No. of Passengers</td>
        <td><input type="number" name="passengers"></td>
    </tr>
    <tr>
        <td>Name of Passenger</td>
        <td><input type="text" name="passenger_name"></td>
    </tr>
    <tr>
        <td>Gender of Passenger</td>
        <td>
            <input type="radio" name="gender"> Male
            <input type="radio" name="gender"> Female
        </td>
    </tr>
    <tr>
        <td colspan="2" align="center">
            <input type="submit" value="Book Ticket">
            <input type="reset" value="Reset">
        </td>
    </tr>
</table>
</form>
</body>
```

```
</html>
```

5. **Design an HTML form for customer registration visiting a departmental store. The form should consist of fields such as name, contact no, gender, preferred days of purchasing, favorite item(to be selected from a list of items),suggestions etc.. You should provide a button to submit as well as reset the form contents.**

```
<!DOCTYPE html>
<html>
<head>
    <title>Customer Registration Form</title>
    <style>
        body {
            font-family: Arial, sans-serif;
        }
        table {
            margin: auto;
            border-collapse: collapse;
        }
        td {
            padding: 8px;
        }
        h2 {
            text-align: center;
        }
    </style>
</head>
```

```
<body>
<h2>Departmental Store Customer Registration</h2>
<form>
<table border="1">
    <tr>
        <td>Name</td>
        <td><input type="text" name="name"></td>
    </tr>
    <tr>
        <td>Contact No</td>
        <td><input type="tel" name="contact"></td>
    </tr>
    <tr>
        <td>Gender</td>
        <td>
            <input type="radio" name="gender"> Male
            <input type="radio" name="gender"> Female
            <input type="radio" name="gender"> Other
        </td>
    </tr>
    <tr>
        <td>Preferred Days of Purchasing</td>
        <td>
```

```
        <input type="checkbox" name="day"> Monday
        <input type="checkbox" name="day"> Tuesday
        <input type="checkbox" name="day"> Wednesday
        <input type="checkbox" name="day"> Thursday
        <input type="checkbox" name="day"> Friday
        <input type="checkbox" name="day"> Saturday
        <input type="checkbox" name="day"> Sunday
    </td>
</tr>
<tr>
    <td>Favorite Item</td>
    <td>
        <select name="item">
            <option>Select Item</option>
            <option>Groceries</option>
            <option>Clothes</option>
            <option>Electronics</option>
            <option>Stationery</option>
            <option>Cosmetics</option>
        </select>
    </td>
</tr>
<tr>
```

```
            <td>Suggestions</td>
            <td>
                <textarea name="suggestions" rows="4" cols="25"></textarea>
            </td>
        </tr>
        <tr>
            <td colspan="2" align="center">
                <input type="submit" value="Submit">
                <input type="reset" value="Reset">
            </td>
        </tr>
    </table>
    </form>
    </body>
    </html>
```

6. **Design a webpage for an Employee Profile, where the second column contains the employee details.**

- **Employee ID**
- **Name**
- **Designation**
- **Department**
- **Email**
- **Photo**

In the first column, provide hyperlinks for Home, About Us, and Contact HR. Extend the above question: If the user clicks on the Contact HR link, display the HR department, email and phone number. Create a form to accept new employee details, including:

- **Name, Employee ID, Designation, Department, Email, and Phone Number.**

```
<!DOCTYPE html>

<html>

<head>

    <title>Employee Profile</title>



    <style>

        body{

            font-family: Arial, sans-serif;

        }

        .container{

            width:80%;

            margin:auto;

            display:flex;

            border:1px solid black;

        }

        .menu{

            width:25%;

            background:#f2f2f2;

            padding:20px;

        }
```

```
.content{
    width:75%;
    padding:20px;
}
a{
    display:block;
    margin:10px 0;
    font-weight:bold;
    text-decoration:none;
    color:blue;
    cursor:pointer;
}
img{
    width:150px;
    height:150px;
    border:1px solid black;
}
table{
    border-collapse:collapse;
    margin-top:20px;
}
td{
    padding:8px;
```

```
        }
        .hidden{
            display:none;
        }
    </style>
    <script>
        function showHR(){
            document.getElementById("hrdetails").style.display="block";
        }
    </script>
</head>
<body>
<h2 align="center">Employee Profile Page</h2>
<div class="container">
    <!-- Column 1 -->
    <div class="menu">
        <h3>Menu</h3>
        <a href="#">Home</a>
        <a href="#">About Us</a>
        <a onclick="showHR()">Contact HR</a>
    </div>
    <!-- Column 2 -->
    <div class="content">
```

```
<h3>Employee Details</h3>
<p><b>Employee ID:</b> EMP101</p>
<p><b>Name:</b> Rahul Sharma</p>
<p><b>Designation:</b> Software Developer</p>
<p><b>Department:</b> IT</p>
<p><b>Email:</b> rahul@gmail.com</p>
<p><b>Photo:</b></p>
<img src="employee.jpg" alt="Employee Photo">
<!-- HR Details -->
<div id="hrdetails" class="hidden">
    <h3>HR Contact Details</h3>
    <p><b>Department:</b> Human Resource</p>
    <p><b>Email:</b> hr@company.com</p>
    <p><b>Phone:</b> 9876543210</p>
</div>
<!-- New Employee Form -->
<h3>Add New Employee</h3>
<form>
<table border="1">
    <tr>
        <td>Name</td>
        <td><input type="text"></td>
    </tr>
```

```
<tr>
    <td>Employee ID</td>
    <td><input type="text"></td>
</tr>
<tr>
    <td>Designation</td>
    <td><input type="text"></td>
</tr>
<tr>
    <td>Department</td>
    <td><input type="text"></td>
</tr>
<tr>
    <td>Email</td>
    <td><input type="email"></td>
</tr>
<tr>
    <td>Phone Number</td>
    <td><input type="tel"></td>
</tr>
<tr>
    <td colspan="2" align="center">
        <input type="submit" value="Submit">
```

```
                <input type="reset" value="Reset">
            </td>
        </tr>
    </table>
    </form>
  </div>
</div>
</body>
</html>
```

7. Create a feedback form with the following fields: Name Email Feedback Message A rating (1–5 stars) Upon form submission, display the user's name, email, and feedback along with a dynamically generated star rating (e.g., if the rating is 4, display 4 stars).

```
<!DOCTYPE html>
<html>
<head>
    <title>Feedback Form</title>
</head>
<body>

<h2>Feedback Form</h2>

<form onsubmit="return showFeedback()">
    Name:<br>
    <input type="text" id="name" required><br><br>
```

```
    Email:<br>
    <input type="email" id="email" required><br><br>

    Feedback Message:<br>
    <textarea id="message" required></textarea><br><br>

    Rating (1–5):<br>
    <select id="rating">
        <option value="1">1 Star</option>
        <option value="2">2 Stars</option>
        <option value="3">3 Stars</option>
        <option value="4">4 Stars</option>
        <option value="5">5 Stars</option>
    </select><br><br>

    <input type="submit" value="Submit">
</form>

<hr>

<div id="output" style="display:none;">
    <h3>Submitted Feedback</h3>
```

```
    <p><b>Name:</b> <span id="outName"></span></p>
    <p><b>Email:</b> <span id="outEmail"></span></p>
    <p><b>Feedback:</b> <span id="outMessage"></span></p>
    <p><b>Rating:</b> <span id="outStars"></span></p>
</div>

<script>
function showFeedback() {
    var name = document.getElementById("name").value;
    var email = document.getElementById("email").value;
    var message = document.getElementById("message").value;
    var rating = document.getElementById("rating").value;

    var stars = "";
    for (var i = 0; i < rating; i++) {
        stars += "*";
    }

    document.getElementById("outName").innerHTML = name;
    document.getElementById("outEmail").innerHTML = email;
    document.getElementById("outMessage").innerHTML = message;
    document.getElementById("outStars").innerHTML = stars;
```

```
        document.getElementById("output").style.display = "block";


        return false; // prevent page refresh

    }

    </script>


    </body>

    </html>
```

PHP Programming

1. Write a PHP program to write a program for find the given number is palindrome Or not.

```
<?php

$number = 121;   // Change number here
$original = $number;
$reverse = 0;

while($number > 0){
   $rem = $number % 10;
   $reverse = ($reverse * 10) + $rem;
   $number = (int)($number / 10);
}

if($original == $reverse){
   echo "Number is Palindrome";
}else{
   echo "Number is Not Palindrome";
}

?>
```

2. Write a PHP script to accept username and password. If in the first three chances, username and password entered is correct then display second form with “Welcome message” otherwise display error message. [Use Session]

login.php

```
<?php
session_start();

// Set default attempts
if(!isset($_SESSION['attempt'])){
    $_SESSION['attempt'] = 0;
}

$message = "";

if(isset($_POST['login'])){

    $username = $_POST['username'];
    $password = $_POST['password'];

    // Set Correct Credentials
    $correct_user = "admin";
    $correct_pass = "1234";

    if($username == $correct_user && $password == $correct_pass){
        $_SESSION['login'] = true;
        header("Location: welcome.php");
        exit();
    }else{
        $_SESSION['attempt']++;

        if($_SESSION['attempt'] >= 3){
            $message = "Maximum login attempts reached!";
        }else{
            $message = "Invalid Username or Password";
        }
    }
}
?>

<html>
<head>
<title>Login</title>
</head>
```

```
<body>

<h2>Login Form</h2>

<?php echo $message; ?>

<?php if($_SESSION['attempt'] < 3){ ?>

<form method="post">
Username: <input type="text" name="username" required><br><br>
Password: <input type="password" name="password" required><br><br>
<input type="submit" name="login" value="Login">
</form>

<?php } ?>

</body>
</html>
```

Welcome.php

```
<?php
session_start();

if(!isset($_SESSION['login'])){
    header("Location: login.php");
    exit();
}
?>

<html>
<head>
<title>Welcome</title>
</head>
<body>

<h2>Welcome! Login Successful</h2>

</body>
</html>
```

3. Write a PHP script to accept Employee details (Eno, Ename, Address) on first page. On second page accept earning (Basic, DA, HRA). On third page print Employee information (Eno, Ename,Address,Basic, DA, HRA, Total)

Emp1.php

```
<?php
session_start();

if(isset($_POST['next'])){
   $_SESSION['eno'] = $_POST['eno'];
   $_SESSION['ename'] = $_POST['ename'];
   $_SESSION['address'] = $_POST['address'];

   header("Location: emp2.php");
}
?>

<html>
<body>

<h2>Employee Details</h2>

<form method="post">
Employee No: <input type="text" name="eno" required><br><br>
Employee Name: <input type="text" name="ename" required><br><br>
Address: <textarea name="address" required></textarea><br><br>
<input type="submit" name="next" value="Next">
</form>
</body>
</html>
```

Emp2.php

```
<?php
session_start();

if(isset($_POST['next'])){
   $_SESSION['basic'] = $_POST['basic'];
   $_SESSION['da'] = $_POST['da'];
   $_SESSION['hra'] = $_POST['hra'];

   header("Location: emp3.php");
```

```
}
?>

<html>
<body>

<h2>Employee Earnings</h2>

<form method="post">
Basic: <input type="number" name="basic" required><br><br>
DA: <input type="number" name="da" required><br><br>
HRA: <input type="number" name="hra" required><br><br>

<input type="submit" name="next" value="Next">
</form>

</body>
</html>
```

Emp3.php

```
<?php
session_start();

$eno = $_SESSION['eno'];
$ename = $_SESSION['ename'];
$address = $_SESSION['address'];
$basic = $_SESSION['basic'];
$da = $_SESSION['da'];
$hra = $_SESSION['hra'];

$total = $basic + $da + $hra;
?>

<html>
<body>

<h2>Employee Information</h2>

Employee No: <?php echo $eno; ?><br><br>
Employee Name: <?php echo $ename; ?><br><br>
```

```
Address: <?php echo $address; ?><br><br>

Basic: <?php echo $basic; ?><br><br>
DA: <?php echo $da; ?><br><br>
HRA: <?php echo $hra; ?><br><br>

<b>Total Salary: <?php echo $total; ?></b>

</body>
</html>
```

4.Design HTML form to accept dimension of a cylinder and write a PHP script to Calculate the area and volume of Cylinder using a function.

Cylinder.html

```
<html>
<head>
   <title>Cylinder Form</title>
</head>
<body>
<h2>Enter Dimensions of Cylinder</h2>
<form method="post" action="cylinder.php">
   Radius: <input type="text" name="radius" required><br><br>
   Height: <input type="text" name="height" required><br><br>
   <input type="submit" value="Calculate">
</form>
</body>
</html>
```

Cylinder.php

```
<?php

function calculateCylinder($r, $h)
{
   $area = 2 * pi() * $r * ($r + $h);   // Surface Area
   $volume = pi() * $r * $r * $h;       // Volume

   return array($area, $volume);
}

$radius = $_POST['radius'];
$height = $_POST['height'];
```

```
list($area, $volume) = calculateCylinder($radius, $height);

echo "<h2>Cylinder Results</h2>";
echo "Radius: " . $radius . "<br><br>";
echo "Height: " . $height . "<br><br>";
echo "Surface Area: " . $area . "<br><br>";
echo "Volume: " . $volume;

?>
```

5. Design HTML form to accept two strings and following three options (use radio button) and write a PHP script to perform the selected operations on String.

String.html

```
<html>
<head>
   <title>String Operations</title>
</head>
<body>

<h2>Enter Two Strings</h2>

<form method="post" action="string.php">

   String 1: <input type="text" name="str1" required><br><br>
   String 2: <input type="text" name="str2" required><br><br>

   <h3>Select Operation:</h3>

   <input type="radio" name="operation" value="concat" required> Concatenate<br>
   <input type="radio" name="operation" value="compare"> Compare Strings<br>
   <input type="radio" name="operation" value="length"> Find Length<br><br>

   <input type="submit" value="Perform Operation">

</form>

</body>
</html>
```

String.php

```php
<?php

$str1 = $_POST['str1'];
$str2 = $_POST['str2'];
$operation = $_POST['operation'];

echo "<h2>Result</h2>";

switch($operation)
{
   case "concat":
      $result = $str1 . " " . $str2;
      echo "Concatenated String: " . $result;
      break;

   case "compare":
      if($str1 == $str2)
         echo "Both Strings are Equal";
      else
         echo "Strings are Not Equal";
      break;

   case "length":
      echo "Length of String 1: " . strlen($str1) . "<br>";
      echo "Length of String 2: " . strlen($str2);
      break;

   default:
      echo "Invalid Operation";
}

?>
```

6. Design a HTML form to accept a number and write a PHP script to display each digit of a number in words.

```html
<!DOCTYPE html>

<html>

<head>
```

```
    <title>Digit in Words</title>
</head>
<body>
<h2>Digits in Words</h2>
<?php
if(isset($_POST['number'])) {
    $number = $_POST['number']
    $digits = str_split($number);
    $words = array(
        "0" => "Zero",
        "1" => "One",
        "2" => "Two",
        "3" => "Three",
        "4" => "Four",
        "5" => "Five",
        "6" => "Six",
        "7" => "Seven",
        "8" => "Eight",
        "9" => "Nine"
    );
    echo "<strong>Entered Number:</strong> " . $number . "<br><br>";
    echo "<strong>Digits in Words:</strong><br>";
    foreach($digits as $digit) {
```

```
        if(array_key_exists($digit, $words)) {
            echo $words[$digit] . " ";
        }
    }
}
?>

</body>
</html>
```

6.Create a PHP program where the user can select items from a product list (e.g., books, pens, or bags) and input quantities.

Products.html

```
<html>
<head>
    <title>Product Selection</title>
</head>
<body>
<h2>Select Products</h2>
<form method="post" action="bill.php">
    <h3>Books (Price: 200)</h3>
    Quantity: <input type="number" name="book_qty" min="0" value="0"><br><br>
    <h3>Pens (Price: 10)</h3>
    Quantity: <input type="number" name="pen_qty" min="0" value="0"><br><br>
```

```
    <h3>Bags (Price: 500)</h3>
    Quantity: <input type="number" name="bag_qty" min="0" value="0"><br><br>
    <input type="submit" value="Generate Bill">
</form>
</body>
</html>
```

Bill.php

```
<?php
// Prices
$book_price = 200;
$pen_price = 10;
$bag_price = 500;
// Quantities from form
$book_qty = $_POST['book_qty'];
$pen_qty = $_POST['pen_qty'];
$bag_qty = $_POST['bag_qty'];
// Calculate totals
$book_total = $book_qty * $book_price;
$pen_total = $pen_qty * $pen_price;
$bag_total = $bag_qty * $bag_price;
$grand_total = $book_total + $pen_total + $bag_total;
echo "<h2>Bill Details</h2>";
echo "Books: $book_qty × 200 = $book_total <br><br>";
```

```
echo "Pens: $pen_qty × 10 = $pen_total <br><br>";

echo "Bags: $bag_qty × 500 = $bag_total <br><br>";

echo "<h3>Grand Total: $grand_total</h3>";

?>
```

7.Design HTML form to accept a number and write a PHP script to calculate the factorial of a number using function.

Factorial.html

```
<html>

<head>

    <title>Factorial Program</title>

</head>

<body>

<h2>Enter a Number</h2>

<form method="post" action="factorial.php">

    Number: <input type="number" name="num" min="0" required>

    <input type="submit" value="Calculate">

</form>

</body>

</html>
```

factorial.php

```
<?php

function factorial($n)

{

    if($n == 0 || $n == 1)
```

```
        return 1;
    else
        return $n * factorial($n - 1);   // Recursive function
}
$number = $_POST['num'];
$result = factorial($number);
echo "<h2>Factorial Result</h2>";
echo "Number: " . $number . "<br><br>";
echo "Factorial: " . $result;
?>
```

8. Design an application to accept username and password. While accepting password check whether is it made of numbers, alphabets, special symbols and password length should be greater than 8. If password is as per requirement, then print “You enter red strong password” else print error message “Enter complex password”.

Login.html

```
<html>
<head>
    <title>Password Validation</title>
</head>
<body>
<h2>Login Form</h2>
<form method="post" action="validate.php">
    Username: <input type="text" name="username" required><br><br>
    Password: <input type="password" name="password" required><br><br>
```

```
    <input type="submit" value="Submit">
</form>
</body>
</html>
```

Validate.php

```
<?php
$username = $_POST['username'];
$password = $_POST['password'];
// Conditions
$length = strlen($password) > 8;
$hasUpper = preg_match("/[A-Z]/", $password);
$hasLower = preg_match("/[a-z]/", $password);
$hasNumber = preg_match("/[0-9]/", $password);
$hasSpecial = preg_match("/[^a-zA-Z0-9]/", $password);
echo "<h2>Password Validation Result</h2>";
if($length && $hasUpper && $hasLower && $hasNumber && $hasSpecial)
{
    echo "<span style='color:red;'>You enter red strong password</span>";
}
else
{
    echo "<span style='color:black;'>Enter complex password</span>";
}
```

```
?>
```

9. Design HTML form to accept book details (bno, bname and price) from user. Write a PHP script to store input information in MySql book table.

Run this in phpMyAdmin / MySQL:

```
CREATE DATABASE bookstore;

CREATE TABLE book (

   bno INT PRIMARY KEY,

   bname VARCHAR(50),

   price INT

);
```

Book.html

```
<html>

<head>

   <title>Book Form</title>

</head>

<body>

<h2>Enter Book Details</h2>

<form method="post" action="insert.php">

   Book No: <input type="number" name="bno" required><br><br>

   Book Name: <input type="text" name="bname" required><br><br>

   Price: <input type="number" step="0.01" name="price" required><br><br>

   <input type="submit" value="Save Book">

</form>

</body>
```

```
</html>
```

Insert.php

```
<?php
// Database connection
$conn = mysqli_connect("localhost", "root", "", "bookstore");
// Get form data
$bno = $_POST['bno'];
$bname = $_POST['bname'];
$price = $_POST['price'];
// Insert query
$sql = "INSERT INTO book (bno, bname, price)
        VALUES ('$bno', '$bname', '$price')";
if(mysqli_query($conn, $sql))
{
    echo "<h3>Book record inserted successfully!</h3>";
}
else
{
    echo "Error: " . mysqli_error($conn);
}
mysqli_close($conn);
?>
```

10.Design an application that accept the full name, contact number, email-id, password and confirm password. If password and confirm password matched the store accept data into database while storing password encrypt it.

Run in phpMyAdmin / MySQL:

```
CREATE DATABASE userdb;

CREATE TABLE users (

   id INT AUTO_INCREMENT PRIMARY KEY,

   fullname VARCHAR(100),

   contact VARCHAR(15),

   email VARCHAR(100),

   password VARCHAR(255)

);
```

Register.html

```
<html>

<head>

   <title>User Registration</title>

</head>

<body>

<h2>User Registration Form</h2>


<form method="post" action="register.php">

Full Name:

<input type="text" name="fullname" required>

<br><br>
```

```
Contact No:

<input type="text" name="contact" required>

<br><br>

Email:

<input type="email" name="email" required>

<br><br>

Password:

<input type="password" name="password" required>

<br><br>

Confirm Password:

<input type="password" name="confirm_password" required>

<br><br>

<input type="submit" name="submit" value="Register">


</form>

</body>

</html>
```

register.php

```
<?php

// Database connection

$conn = mysqli_connect("localhost", "root", "", "userdb");

// Check connection

if (!$conn) {
```

```
    echo "Connection Failed: " . mysqli_connect_error();
}
// Check form submit
if (isset($_POST['submit'])) {
    // Get form data
    $fullname = $_POST['fullname'];
    $contact = $_POST['contact'];
    $email = $_POST['email'];
    $password = $_POST['password'];
    $confirm_password = $_POST['confirm_password'];
    // Check password match
    if ($password != $confirm_password) {
        echo "Password and Confirm Password do not match!";
        exit();
    }
    // Encrypt password
    $encrypted_password = password_hash($password, PASSWORD_DEFAULT);
    // Insert query
    $sql = "INSERT INTO users (fullname, contact, email, password)
            VALUES ('$fullname', '$contact', '$email', '$encrypted_password')";
    // Execute query
    if (mysqli_query($conn, $sql)) {
        echo "Registration Successful!";
```

```
    } else {
        echo "Error: " . mysqli_error($conn);
    }
}
// Close connection
mysqli_close($conn);
?>
```

XML

1. **Write a script to create XML file named “Subject.xml” Stored at least 3 records containing subject id, subject name, class, semester etc.**

```
<?php
// Create XML document
$xml = new DOMDocument("1.0", "UTF-8");
$xml->formatOutput = true;
// Create root element
$subjects = $xml->createElement("Subjects");
$xml->appendChild($subjects);
// Subject 1
$subject1 = $xml->createElement("Subject");
$subject1->appendChild($xml->createElement("SubjectID", "101"));
$subject1->appendChild($xml->createElement("SubjectName", "Database
Management System"));
$subject1->appendChild($xml->createElement("Class", "TYCDS"));
```

```
$subject1->appendChild($xml->createElement("Semester", "5"));

$subjects->appendChild($subject1);

// Subject 2

$subject2 = $xml->createElement("Subject");

$subject2->appendChild($xml->createElement("SubjectID", "102"));

$subject2->appendChild($xml->createElement("SubjectName", "Computer
Networks"));

$subject2->appendChild($xml->createElement("Class", "TYCDS"));

$subject2->appendChild($xml->createElement("Semester", "5"));

$subjects->appendChild($subject2);

// Subject 3

$subject3 = $xml->createElement("Subject");

$subject3->appendChild($xml->createElement("SubjectID", "103"));

$subject3->appendChild($xml->createElement("SubjectName", "Web
Technology"));

$subject3->appendChild($xml->createElement("Class", "TYCDS"));

$subject3->appendChild($xml->createElement("Semester", "6"));

$subjects->appendChild($subject3);

// Save XML file

$xml->save("Subject.xml");

echo "Subject.xml file created successfully!";

?>
```

```
<?xml version="1.0" encoding="UTF-8"?>

<Subjects>
```

```
  <Subject>
    <SubjectID>101</SubjectID>
    <SubjectName>Database Management System</SubjectName>
    <Class>TYCDS</Class>
    <Semester>5</Semester>
  </Subject>
  <Subject>
    <SubjectID>102</SubjectID>
    <SubjectName>Computer Networks</SubjectName>
    <Class>TYCDS</Class>
    <Semester>5</Semester>
  </Subject>
  <Subject>
    <SubjectID>103</SubjectID>
    <SubjectName>Web Technology</SubjectName>
    <Class>TYCDS</Class>
    <Semester>6</Semester>
  </Subject>
</Subjects>
```

2. **Write a script to create “cricket.xml” file with multiple elements as shown below:**

   <Cricket Team>

   <Team Country =“Australia”>

   <Player>----------</Player>

   <Runs>----------</Runs>

   <Wicket>----------</Wicket.

   </Team>

   </Create Team>

   Write a script to add multiple elements in “cricket.xml” file of category ,”country=India”

   cricket.php

```
<?php

$xml = new DOMDocument("1.0", "UTF-8");

$xml->formatOutput = true;



// Root element

$cricket = $xml->createElement("CricketTeam");

$xml->appendChild($cricket);



// -------- Team Australia --------

$team1 = $xml->createElement("Team");

$team1->setAttribute("Country", "Australia");
```

```
$team1->appendChild($xml->createElement("Player", "David Warner"));
$team1->appendChild($xml->createElement("Runs", "85"));
$team1->appendChild($xml->createElement("Wicket", "0"));

$cricket->appendChild($team1);

// -------- Team England --------
$team2 = $xml->createElement("Team");
$team2->setAttribute("Country", "England");

$team2->appendChild($xml->createElement("Player", "Joe Root"));
$team2->appendChild($xml->createElement("Runs", "72"));
$team2->appendChild($xml->createElement("Wicket", "1"));

$cricket->appendChild($team2);

// Save file
$xml->save("cricket.xml");

echo "cricket.xml file created successfully!";
?>
```

addIndia.php

```
<?php
```

```
$xml = new DOMDocument();
$xml->load("cricket.xml");

// Get root element
$cricket = $xml->getElementsByTagName("CricketTeam")->item(0);

// -------- Team India --------
$teamIndia = $xml->createElement("Team");
$teamIndia->setAttribute("Country", "India");

$teamIndia->appendChild($xml->createElement("Player", "Virat Kohli"));
$teamIndia->appendChild($xml->createElement("Runs", "95"));
$teamIndia->appendChild($xml->createElement("Wicket", "0"));

$cricket->appendChild($teamIndia);

// Save updated file
$xml->save("cricket.xml");

echo "India team added successfully!";
?>
```

3. **Create a XML document which contains details of cars car like: id, company name, model engine and mileage and display the same as a table by using element by using XSLT.**

Cars.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<?xml-stylesheet type="text/xsl" href="cars.xsl"?>

<Cars>

  <Car>

    <Id>1</Id>

    <Company>Toyota</Company>

    <Model>Fortuner</Model>

    <Engine>2.8L Diesel</Engine>

    <Mileage>14 kmpl</Mileage>

  </Car>

  <Car>

    <Id>2</Id>

    <Company>Honda</Company>

    <Model>City</Model>

    <Engine>1.5L Petrol</Engine>

    <Mileage>18 kmpl</Mileage>

  </Car>

  <Car>

    <Id>3</Id>

    <Company>Hyundai</Company>
```

```
        <Model>Creta</Model>
        <Engine>1.5L Diesel</Engine>
        <Mileage>19 kmpl</Mileage>
    </Car>
</Cars>
```

Cars.xsl

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
<xsl:template match="/">
<html>
<head>
    <title>Car Details</title>
</head>
<body>
<h2>Car Information Table</h2>
<table border="1">
    <tr bgcolor="lightblue">
        <th>ID</th>
        <th>Company</th>
        <th>Model</th>
        <th>Engine</th>
        <th>Mileage</th>
```

```
    </tr>
    <xsl:for-each select="Cars/Car">
        <tr>
            <td><xsl:value-of select="Id"/></td>
            <td><xsl:value-of select="Company"/></td>
            <td><xsl:value-of select="Model"/></td>
            <td><xsl:value-of select="Engine"/></td>
            <td><xsl:value-of select="Mileage"/></td>
        </tr>
    </xsl:for-each>
  </table>
  </body>
  </html>
  </xsl:template>
  </xsl:stylesheet>
```

4. **Write a script that generates an XML file named “Student.xml” containing at least 5 student records. Each record should include the following data:**

   - **Student ID**
   - **Student Name**
   - **Age**
   - **Gender**
   - **Program**

```
<?php
// Create XML Document
```

```
$xml = new DOMDocument("1.0", "UTF-8");

$xml->formatOutput = true;

// Create root element

$students = $xml->createElement("Students");

$xml->appendChild($students);

// -------- Student 1 --------

$student1 = $xml->createElement("Student");

$student1->appendChild($xml->createElement("StudentID", "101"));

$student1->appendChild($xml->createElement("StudentName", "Rahul
Sharma"));

$student1->appendChild($xml->createElement("Age", "20"));

$student1->appendChild($xml->createElement("Gender", "Male"));

$student1->appendChild($xml->createElement("Program", "BCA"));

$students->appendChild($student1);

// -------- Student 2 --------

$student2 = $xml->createElement("Student");

$student2->appendChild($xml->createElement("StudentID", "102"));

$student2->appendChild($xml->createElement("StudentName", "Sneha Patil"));

$student2->appendChild($xml->createElement("Age", "19"));

$student2->appendChild($xml->createElement("Gender", "Female"));

$student2->appendChild($xml->createElement("Program", "BSc IT"));

$students->appendChild($student2);

// -------- Student 3 --------

$student3 = $xml->createElement("Student");
```

```
$student3->appendChild($xml->createElement("StudentID", "103"));

$student3->appendChild($xml->createElement("StudentName", "Amit Joshi"));

$student3->appendChild($xml->createElement("Age", "21"));

$student3->appendChild($xml->createElement("Gender", "Male"));

$student3->appendChild($xml->createElement("Program", "BCA"));

$students->appendChild($student3);

// -------- Student 4 --------

$student4 = $xml->createElement("Student");

$student4->appendChild($xml->createElement("StudentID", "104"));

$student4->appendChild($xml->createElement("StudentName", "Priya Desai"));

$student4->appendChild($xml->createElement("Age", "22"));

$student4->appendChild($xml->createElement("Gender", "Female"));

$student4->appendChild($xml->createElement("Program", "BSc CS"));

$students->appendChild($student4);

// -------- Student 5 --------

$student5 = $xml->createElement("Student");

$student5->appendChild($xml->createElement("StudentID", "105"));

$student5->appendChild($xml->createElement("StudentName", "Rohan
Kulkarni"));

$student5->appendChild($xml->createElement("Age", "20"));

$student5->appendChild($xml->createElement("Gender", "Male"));

$student5->appendChild($xml->createElement("Program", "BCA"));

$students->appendChild($student5);

// Save XML File
```

```
$xml->save("Student.xml");

echo "Student.xml file created successfully!";

?>
```

5. **Write a script to create an XML file called “Employees.xml”containing records for 5 employees. Each record should contain:**

   - **Employee ID**
   - **Name**
   - **Position**
   - **Department**
   - **Date of Joining**

```
<?php

// Create XML document

$xml = new DOMDocument("1.0", "UTF-8");

$xml->formatOutput = true;

// Create root element

$employees = $xml->createElement("Employees");

$xml->appendChild($employees);

// -------- Employee 1 --------

$emp1 = $xml->createElement("Employee");

$emp1->appendChild($xml->createElement("EmployeeID", "E101"));

$emp1->appendChild($xml->createElement("Name", "Amit Sharma"));

$emp1->appendChild($xml->createElement("Position", "Manager"));

$emp1->appendChild($xml->createElement("Department", "Sales"));

$emp1->appendChild($xml->createElement("DateOfJoining", "2019-06-15"));
```

```
$employees->appendChild($emp1);
// -------- Employee 2 --------
$emp2 = $xml->createElement("Employee");
$emp2->appendChild($xml->createElement("EmployeeID", "E102"));
$emp2->appendChild($xml->createElement("Name", "Sneha Patil"));
$emp2->appendChild($xml->createElement("Position", "Developer"));
$emp2->appendChild($xml->createElement("Department", "IT"));
$emp2->appendChild($xml->createElement("DateOfJoining", "2020-08-10"));
$employees->appendChild($emp2);
// -------- Employee 3 --------
$emp3 = $xml->createElement("Employee");
$emp3->appendChild($xml->createElement("EmployeeID", "E103"));
$emp3->appendChild($xml->createElement("Name", "Rohit Verma"));
$emp3->appendChild($xml->createElement("Position", "Accountant"));
$emp3->appendChild($xml->createElement("Department", "Finance"));
$emp3->appendChild($xml->createElement("DateOfJoining", "2018-03-25"));
$employees->appendChild($emp3);
// -------- Employee 4 --------
$emp4 = $xml->createElement("Employee");
$emp4->appendChild($xml->createElement("EmployeeID", "E104"));
$emp4->appendChild($xml->createElement("Name", "Priya Desai"));
$emp4->appendChild($xml->createElement("Position", "HR Executive"));
$emp4->appendChild($xml->createElement("Department", "HR"));
```

```
$emp4->appendChild($xml->createElement("DateOfJoining", "2021-01-05"));
$employees->appendChild($emp4);
// -------- Employee 5 --------
$emp5 = $xml->createElement("Employee");
$emp5->appendChild($xml->createElement("EmployeeID", "E105"));
$emp5->appendChild($xml->createElement("Name", "Karan Mehta"));
$emp5->appendChild($xml->createElement("Position", "System Analyst"));
$emp5->appendChild($xml->createElement("Department", "IT"));
$emp5->appendChild($xml->createElement("DateOfJoining", "2017-11-12"));
$employees->appendChild($emp5);
// Save XML file
$xml->save("Employees.xml");
echo "Employees.xml file created successfully!";
?>
```

6. **Write a script to create an XML file called “Timetable.xml” containing 3 records for a weekly class timetable. Each record should include:**

   - **Day of the Week**
   - **Class Name**
   - **Instructor**
   - **Time Slot**
   - **Classroom**

```
<?php
// Create XML document
$xml = new DOMDocument("1.0", "UTF-8");
```

```
$xml->formatOutput = true;

// Create root element

$timetable = $xml->createElement("Timetable");

$xml->appendChild($timetable);

// -------- Record 1 --------

$record1 = $xml->createElement("Schedule");

$record1->appendChild($xml->createElement("DayOfWeek", "Monday"));

$record1->appendChild($xml->createElement("ClassName", "Database
Management System"));

$record1->appendChild($xml->createElement("Instructor", "Dr. Sharma"));

$record1->appendChild($xml->createElement("TimeSlot", "10:00 AM - 11:00
AM"));

$record1->appendChild($xml->createElement("Classroom", "Room 101"));

$timetable->appendChild($record1);

// -------- Record 2 --------

$record2 = $xml->createElement("Schedule");

$record2->appendChild($xml->createElement("DayOfWeek", "Wednesday"));

$record2->appendChild($xml->createElement("ClassName", "Computer
Networks"));

$record2->appendChild($xml->createElement("Instructor", "Prof. Patil"));

$record2->appendChild($xml->createElement("TimeSlot", "11:00 AM - 12:00
PM"));

$record2->appendChild($xml->createElement("Classroom", "Room 202"));

$timetable->appendChild($record2);

// -------- Record 3 --------
```

```
$record3 = $xml->createElement("Schedule");

$record3->appendChild($xml->createElement("DayOfWeek", "Friday"));

$record3->appendChild($xml->createElement("ClassName", "Web
Technology"));

$record3->appendChild($xml->createElement("Instructor", "Ms. Desai"));

$record3->appendChild($xml->createElement("TimeSlot", "1:00 PM - 2:00 PM"));

$record3->appendChild($xml->createElement("Classroom", "Room 303"));

$timetable->appendChild($record3);

// Save XML file

$xml->save("Timetable.xml");

echo "Timetable.xml file created successfully!";

?>
```

7. Write a script to generate an XML file called &”Course Registrations.xml”; that stores 5 records of students registering for courses. Each record should contain:

- **Registration ID**
- **Student Name**
- **Course Name**
- **Instructor**
- **Date of Registration**

```
<?php

// Create XML document

$xml = new DOMDocument("1.0", "UTF-8");

$xml->formatOutput = true;
```

```
// Root element
$registrations = $xml->createElement("CourseRegistrations");
$xml->appendChild($registrations);

// -------- Record 1 --------
$reg1 = $xml->createElement("Registration");
$reg1->appendChild($xml->createElement("RegistrationID", "R101"));
$reg1->appendChild($xml->createElement("StudentName", "Rahul Sharma"));
$reg1->appendChild($xml->createElement("CourseName", "Web Development"));
$reg1->appendChild($xml->createElement("Instructor", "Prof. Patil"));
$reg1->appendChild($xml->createElement("DateOfRegistration", "2024-01-15"));
$registrations->appendChild($reg1);

// -------- Record 2 --------
$reg2 = $xml->createElement("Registration");
$reg2->appendChild($xml->createElement("RegistrationID", "R102"));
$reg2->appendChild($xml->createElement("StudentName", "Sneha Desai"));
$reg2->appendChild($xml->createElement("CourseName", "Database Systems"));
$reg2->appendChild($xml->createElement("Instructor", "Dr. Mehta"));
$reg2->appendChild($xml->createElement("DateOfRegistration", "2024-01-16"));
$registrations->appendChild($reg2);
// -------- Record 3 -------
$reg3 = $xml->createElement("Registration");
```

```
$reg3->appendChild($xml->createElement("RegistrationID", "R103"));
$reg3->appendChild($xml->createElement("StudentName", "Amit Joshi"));
$reg3->appendChild($xml->createElement("CourseName", "Computer Networks"));
$reg3->appendChild($xml->createElement("Instructor", "Ms. Sharma"));
$reg3->appendChild($xml->createElement("DateOfRegistration", "2024-01-17"));
$registrations->appendChild($reg3);
// -------- Record 4 --------
$reg4 = $xml->createElement("Registration");
$reg4->appendChild($xml->createElement("RegistrationID", "R104"));
$reg4->appendChild($xml->createElement("StudentName", "Priya Kulkarni"));
$reg4->appendChild($xml->createElement("CourseName", "Software Engineering"));
$reg4->appendChild($xml->createElement("Instructor", "Dr. Rao"));
$reg4->appendChild($xml->createElement("DateOfRegistration", "2024-01-18"));
$registrations->appendChild($reg4);
// -------- Record 5 --------
$reg5 = $xml->createElement("Registration");
$reg5->appendChild($xml->createElement("RegistrationID", "R105"));
$reg5->appendChild($xml->createElement("StudentName", "Rohan Patil"));
$reg5->appendChild($xml->createElement("CourseName", "Artificial Intelligence"));
$reg5->appendChild($xml->createElement("Instructor", "Prof. Deshmukh"));
$reg5->appendChild($xml->createElement("DateOfRegistration", "2024-01-19"));
$registrations->appendChild($reg5);
// Save XML file
```

```
$xml->save("CourseRegistrations.xml");

echo "CourseRegistrations.xml file created successfully!";

?>
```

8. Write a script to create “breakfast.xml” file with multiple elements as shown below: French Fries Rs45 Young youths are very much interested to eat it 650 Write a script to add multiple elements in “breakfast.xml” file of category, Juice.

createBreakfast.php

```
<?php
$xml = new DOMDocument("1.0", "UTF-8");
$xml->formatOutput = true;

// Root element
$breakfast = $xml->createElement("BreakfastMenu");
$xml->appendChild($breakfast);

// -------- Item 1 --------
$item1 = $xml->createElement("Item");
$item1->appendChild($xml->createElement("Name", "French Fries"));
$item1->appendChild($xml->createElement("Price", "Rs45"));
$item1->appendChild($xml->createElement("Description", "Young youths are very much
interested to eat it"));
$item1->appendChild($xml->createElement("Calories", "650"));
$breakfast->appendChild($item1);

// -------- Item 2 --------
$item2 = $xml->createElement("Item");
$item2->appendChild($xml->createElement("Name", "Burger"));
$item2->appendChild($xml->createElement("Price", "Rs80"));
$item2->appendChild($xml->createElement("Description", "Very tasty and popular fast
food"));
$item2->appendChild($xml->createElement("Calories", "700"));
$breakfast->appendChild($item2);

// Save XML file
$xml->save("breakfast.xml");

echo "breakfast.xml file created successfully!";
?>
```

addJuice.php

```php
<?php
$xml = new DOMDocument();
$xml->load("breakfast.xml");

// Get root element
$breakfast = $xml->getElementsByTagName("BreakfastMenu")->item(0);

// -------- Juice 1 --------
$juice1 = $xml->createElement("Juice");
$juice1->appendChild($xml->createElement("Name", "Orange Juice"));
$juice1->appendChild($xml->createElement("Price", "Rs60"));
$juice1->appendChild($xml->createElement("Description", "Fresh and healthy drink"));
$juice1->appendChild($xml->createElement("Calories", "120"));
$breakfast->appendChild($juice1);

// -------- Juice 2 --------
$juice2 = $xml->createElement("Juice");
$juice2->appendChild($xml->createElement("Name", "Mango Juice"));
$juice2->appendChild($xml->createElement("Price", "Rs70"));
$juice2->appendChild($xml->createElement("Description", "Sweet and refreshing"));
$juice2->appendChild($xml->createElement("Calories", "150"));
$breakfast->appendChild($juice2);

// Save updated file
$xml->save("breakfast.xml");

echo "Juice items added successfully!";
?>
```

SQL & Database Design

1. **Create the following relations : Emp(eno,name,dno,salary) Project(pno,pname,controldno,budget)…**

```sql
-- Create Database

CREATE DATABASE CompanyDB;



-- Use Database
```

```
USE CompanyDB;

-- Create Employee Table
CREATE TABLE Emp (
    eno INT PRIMARY KEY,
    name VARCHAR(50),
    dno INT,
    salary DECIMAL(10,2)
);

-- Create Project Table
CREATE TABLE Project (
    pno INT PRIMARY KEY,
    pname VARCHAR(50),
    controldno INT,
    budget DECIMAL(12,2)
);
```

SQL Injection / Security Attack

1. **Using SQL injection attack on Login check whether you can log into another user's account without knowing the correct password.**
2. **Perform SQL injection to modify another user profile data through an UPDATE query**
3. **Show the SQL injection attack on UPDATE statement, you need to make an unauthorized modification to the database by modifying another user's profile.**

Web Services

1. **Write a simple web service called Dairy Product Price Service. Dairy product price table to record the different kinds of product prices. It will supply 3 services to the client so that clients can add product price, delete product price, update product price.**

1. Create Database Table

```
CREATE DATABASE dairy;

USE dairy;

CREATE TABLE dairy_products(

id INT PRIMARY KEY,

product_name VARCHAR(30),

price INT

);
```

2. Simple Web Service (dairy_service.php)

```
<?php

$conn = mysqli_connect("localhost","root","","dairy");

$action = $_GET['action'];

$id = $_GET['id'];

if($action=="add")

{

$name = $_GET['name'];

$price = $_GET['price'];

$sql = "INSERT INTO dairy_products VALUES('$id','$name','$price')";

}

elseif($action=="delete")
```

```
{
$sql = "DELETE FROM dairy_products WHERE id='$id'";
}
elseif($action=="update")
{
$price = $_GET['price'];
$sql = "UPDATE dairy_products SET price='$price' WHERE id='$id'";
}
mysqli_query($conn,$sql);
echo "Operation Successful";
?>
```

3.Client Request

http://localhost/tycds/dairy_service.php?action=add&id=4&name=Paneer&price=300

http://localhost/tycds/dairy_service.php?action=delete&id=4

http://localhost/tycds/dairy_service.php?action=update&id=1&price=60

2. **Create a simple web service called Library Book Service. It should manage a table of books and their details (title, author, genre, and price). The service must provide the following.**

   functionalities:

   Add Book: Add a new book to the table.

   Delete Book: Remove a book based on its ID.

   Update Book: Modify the details of an existing book.

   Get Book Details: Retrieve details of all books or a specific book by ID.…

1. Database and Table Creation

2. Web Service Code (library_service.php)

```
<?php
$conn = mysqli_connect("localhost","root","","library");
$action = $_GET['action'];
$id = $_GET['id'];
if($action=="add")
{
$title = $_GET['title'];
$author = $_GET['author'];
$genre = $_GET['genre'];
$price = $_GET['price'];
$sql = "INSERT INTO books VALUES('$id','$title','$author','$genre','$price')";
}
elseif($action=="delete")
{
$sql = "DELETE FROM books WHERE id='$id'";
}
elseif($action=="update")
{
$title = $_GET['title'];
$author = $_GET['author'];
$genre = $_GET['genre'];
```

```
$price = $_GET['price'];
$sql = "UPDATE books SET
title='$title',
author='$author',
genre='$genre',
price='$price'
WHERE id='$id'";
}
elseif($action=="get")
{
$result = mysqli_query($conn,"SELECT * FROM books");
while($row = mysqli_fetch_assoc($result))
{
echo $row['id']." ".
    $row['title']." ".
    $row['author']." ".
    $row['genre']." ".
    $row['price']."<br>";
}
exit();
}
mysqli_query($conn,$sql);
echo "Operation Successful";
```

```
?>
```

3. Client Requests (Browser)

1.Add Book
http://localhost/library_service.php?action=add&id=3&title=Python&author=Guido&genre=Programming&price=700

2.Delete Book
http://localhost/library_service.php?action=delete&id=3

3. Update Book
http://localhost/library_service.php?action=update&id=1&title=C%20Programming&author=Dennis%20Ritchie&genre=Programming&price=550

4. Get All Book Details
http://localhost/library_service.php?action=get

API

1. Write a program to access online weather API.

Step 1: Get API Key

1. Go to: https://openweathermap.org
2. Create free account
3. Copy your **API Key**

Step 2: Create File weather.php

```
<?php

$apiKey = "YOUR_API_KEY";   // Replace with your API key

$city = "Pune";
```

```
$url =
    "https://api.openweathermap.org/data/2.5/weather?q=$city&appid=$apiKey&units=m
    etric";


$response = file_get_contents($url);


$data = json_decode($response, true);


echo "<h2>Weather Information</h2>";

echo "City: " . $data['name'] . "<br>";

echo "Temperature: " . $data['main']['temp'] . " °C<br>";

echo "Humidity: " . $data['main']['humidity'] . " %<br>";

echo "Weather: " . $data['weather'][0]['description'];

?>
```

How to Run

1. Start **XAMPP / WAMP**
2. Save file inside htdocs
3. Open browser: http://localhost/weather.php

2. **Write a program that uses a currency exchange API (e.g., Exchange Rate-API or Fixer.io) to perform the following:**

   - **Convert a specified amount from one currency to another.**
   - **Display the exchange rate for a given currency pair.**
   - **Handle invalid currency codes or API errors gracefully.**

Step 1: Get Free API Key

Register at: https://www.exchangerate-api.com/

Step 2: Create currency.php

```
<?php

$apiKey = "YOUR_API_KEY";  // Replace with your API key

$from = "USD";

$to = "INR";

$amount = 100;


// API URL

$url = "https://v6.exchangerate-api.com/v6/$apiKey/latest/$from";


// Get API response

$response = file_get_contents($url);

$data = json_decode($response, true);


// Get conversion rate

$rate = $data['conversion_rates'][$to];


// Convert amount

$convertedAmount = $amount * $rate;
```

```
// Display result
echo "<h2>Currency Conversion</h2>";
echo "From: $amount $from <br>";
echo "To: $convertedAmount $to <br>";
echo "Exchange Rate: 1 $from = $rate $to";
?>
```